Instagram User Analytics

Project Description:
The main goal of this project is to analyse and gain insights about "user interaction and engagement" with the app. In this project SQL is used to perform the analysis. The focus is on identifying trends in user activity, user engagement and detecting any suspicious behaviour on the platform. This analysis help in understanding the user behaviour, enabling the marketing team to design new marketing strategies, and providing key metrics that can be valuable to the stakeholders and investors.

Approach:
step 1: created a new database and named the database as ig_clone in MySQL workbench.
step 2: performed data cleaning by checking for null values, removing duplicates and ensuring data consistency.
step 3: used SQL queries to answer all the defined business problems.

Tech-Stack used:
Software and Version: MySQL workbench 8.0.43
Why MySQL workbench: Provides a better interface to write, execute, and visualize queries with tabular output.

Tasks Performed:

a) Marketing analysis:

1. Identify the five oldest users on Instagram from the provided database.

   SQL query:

```sql
# Identify the five oldest users on Instagram from the provided database.
SELECT
    *
FROM
    users
ORDER BY created_at
LIMIT 5;
```

Output:

id	username	created_at
80	Darby_Herzog	2016-05-06 00:14:21
67	Emilio_Bernier52	2016-05-06 13:04:30
63	Elenor88	2016-05-08 01:30:41
95	Nicole71	2016-05-09 17:30:22
38	Jordyn.Jacobson2	2016-05-14 07:56:26
NULL	NULL	NULL

2. Identify users who have never posted a single photo on Instagram.

SQL query:

```
SELECT
    users.id, users.username
FROM
    users
        LEFT JOIN
    photos ON users.id = photos.user_id
WHERE
    photos.user_id IS NULL
ORDER BY username ASC;
```

Output:

id	username
5	Aniya_Hackett
83	Bartholome.Bernhard
91	Bethany20
80	Darby_Herzog
45	David.Osinski47
54	Duane60
90	Esmeralda.Mraz57
81	Esther.Zulauf61
68	Franco_Keebler64
74	Hulda.Macejkovic
14	Jaclyn81
76	Janelle.Nikolaus81
89	Jessyca_West
57	Julien_Schmidt
7	Kasandra_Homenick
75	Leslie67
53	Linnea59
24	Maxwell.Halvorson
41	Mckenna17
66	Mike.Auer39
49	Morgan.Kassulke
71	Nia_Haag
36	Ollie_Ledner37
34	Pearl7
21	Rocio33
25	Tierra.Trantow

3. Determine the winner of the contest and provide their details to the team.

SQL Query:

```
SELECT
    photos.id ,users.username, COUNT(likes.user_id) AS total_like
FROM
    photos
        JOIN
    users ON  photos.user_id = users.id
        LEFT JOIN
    likes ON photos.id = likes.photo_id
GROUP BY photos.id ,users.username
ORDER BY total_like DESC
LIMIT 1;
```

Output:

Result Grid | Filter Rows: | Export:

id	username	total_like
145	Zack_Kemmer93	48

4. Identify and suggest the top five most commonly used hashtags on the platform.

SQL Query:

```
SELECT
    tag_name, COUNT(tags.tag_name) AS common_tags
FROM
    tags
        LEFT JOIN
    photo_tags ON tags.id = photo_tags.tag_id
GROUP BY tags.tag_name
ORDER BY common_tags DESC
LIMIT 5;
```

Output:

Result Grid | Filter Rows: | Export: | Wr

tag_name	common_tags
smile	59
beach	42
party	39
fun	38
concert	24

5. Determine the day of the week when most users register on Instagram.

SQL Query:

```sql
SELECT
    DAYNAME(created_at) AS days, COUNT(*)
FROM
    users
GROUP BY days
ORDER BY COUNT(*) DESC;
```

Output:

days	COUNT(*)
Thursday	16
Sunday	16
Friday	15
Tuesday	14
Monday	14
Wednesday	13
Saturday	12

B) Investor Metrics:

1. a) Calculate the average number of posts per user on Instagram.

SQL Query:

```sql
SELECT round(
    AVG(post),2) AS avg_post
FROM
    (SELECT
        COUNT(photos.id) AS post, users.id
    FROM
        users
    LEFT JOIN photos ON photos.user_id = users.id
    GROUP BY users.id) AS post_cnt;
```

Scalar subquery is used here.

Output:

avg_post
2.57

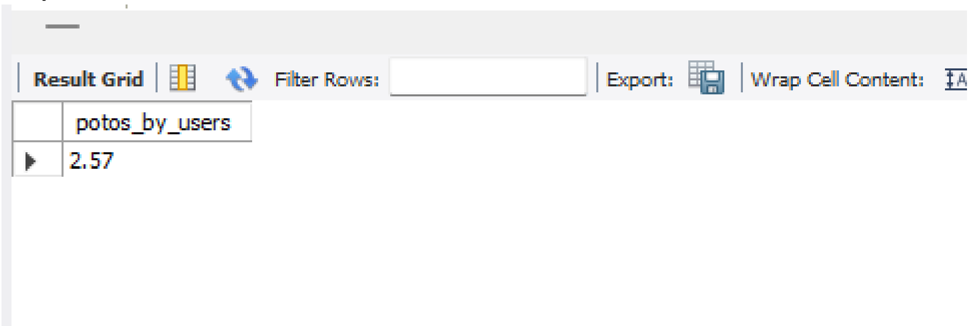
In the above SQL query, I have used round () function to limit the result into 2 decimal places. If we don't use the round () function we will get the average as 2.700

b) Also, provide the total number of photos on Instagram divided by the total number of users.

SQL Query:

```sql
SELECT
    (ROUND((SELECT
                COUNT(*)
            FROM
                photos) / (SELECT
                COUNT(*)
            FROM
                users),
        2)) AS potos_by_users;
```

Output:



We can observe that first and second part of the question are one and the same, (total photos /total users). So, the result to the query is identical.

2. Identify users (potential bots) who have liked every single photo on the site, as this is not typically possible for a normal user.

   SQL Query:

```sql
SELECT
    users.id, users.username
FROM
    users
        JOIN
    likes ON users.id = likes.user_id
GROUP BY users.id , users.username
HAVING count(distinct likes.photo_id) = (SELECT
        count(*)
    FROM
        photos);
```

Output:

id	username
5	Aniya_Hackett
14	Jaclyn81
21	Rocio33
24	Maxwell.Halvorson
36	Ollie_Ledner37
41	Mckenna17
54	Duane60
57	Julien_Schmidt
66	Mike.Auer39
71	Nia_Haag
75	Leslie67
76	Janelle.Nikolaus81
91	Bethany20

- Above output contains the username and id of the bot users.

Insights:

- 26 users have never posted any photo on the platform.

- Winner of the contest – username : Zack_Kemmer93
  User Id : 52
  Total likes : 48

- By the analysis we can determine that smile, beach, party, fun, concert. The brand can use these hashtags to reach the people.

- Most user registration have occurred on Thursday and Sunday (16 each), followed by Friday (15). This indicates the user engagement is high during the weekends and late weekdays. So, the ad campaign should be scheduled around Thursday evening and Sundays for maximum visibility and reach.

- On average, each user has posted 2.57 posts on the platform. With 257 photos across 100 users, the user activity is relatively low. This indicates a potential to improve user engagement and encourage content creation to drive growth.

- On the platform there are 13 accounts that appear to be potential bots, because these accounts have liked all 257 photos.

Monitoring these account is important to maintain the engagement metrics.

Result:

- Gained hands on experience in SQL.
- Learned how to write well Structured SQL queries.
- Identified patterns in user engagement.
- We use can a bot-detection algorithm to identify and remove such account from the platform ensuring organic user engagement.
- So, while analysing the potential bot accounts, I analysed the bot accounts the have liked the post of the contest winner. Out of 48 likes 13 of them were potential bot accounts, leaving 35 genuine likes.

  Query to analyse the bot accounts that have liked the contest winner photo.

```
SELECT
    users.username AS bot_users
FROM
    likes
        JOIN
    users ON likes.user_id = users.id
WHERE
    likes.photo_id = 145
        AND likes.user_id IN (SELECT
            likes.user_id
        FROM
            likes
        GROUP BY likes.user_id
        HAVING COUNT(DISTINCT likes.photo_id) = (SELECT
                COUNT(*)
            FROM
                photos))
```